

Application: Nieuwerburgh.Paul.10025

Paul Nieuwerburgh - paul.nieuwerburgh@gmail.com
Regeneron Science Talent Search

Summary

ID: 2026-5090956530

Task 1: Basic Information

Completed - Jun 15 2025

Basic Information

Please be aware that Evaluators and Judges will see your responses to all questions in Tasks 1-9.

ELIGIBILITY

1. Welcome! Review the [Regeneron STS Official Rules](#) book to make sure you are eligible and the [Ethics Statement](#) to make sure you are in compliance with all rules and expectations.

Select both A & B below after you have read the materials.

A. Official Rules	I have read the Regeneron STS Official Rules
B. Ethics Statement	I have read the Ethics Statement

2. Regeneron STS is an individual competition, seeking to identify future leaders in STEM through a holistic review process. Entrants must only submit individual projects conducted without any support from fellow high school students.

I am submitting INDIVIDUAL research that was not conducted with other high school students.

3. Regeneron STS is a competition for students in their final year of high school.
- Entrants who live and attend school in the United States are eligible to apply, regardless of citizenship.
 - US citizens living abroad are also eligible to apply.
 - Students living abroad who are not U.S. citizens are not eligible and should not apply (but please check out our sister competition, [Regeneron ISEF](#)).

Select the option that best describes you:

I am a high school senior living AND attending school in the United States with an expected graduation date of Winter 2025 or Spring 2026

4. Are you graduating from high school early (or possibly finishing high school early)?

Are you completing your high school coursework in less than 4 years and/or graduating a year or semester early and/or choosing to forgo graduating from high school? We ask this question to verify your eligibility for Regeneron STS.

No

ABOUT YOU

5. Your Legal Name

Enter your name as it appears on your transcript and driver's license (or other government-issued ID if you have one). This information is used to help match certain materials, like your transcript, to your application.

Legal First Name on Your Transcript	Paul
Legal Middle Name on Your Transcript (optional)	Alexander
Legal Last Name on Your Transcript	Nieuwerburgh

6. Your Chosen Name

Enter your name as you prefer to be known. We will use this name to email and address you throughout the application cycle after you have submitted Task 1.

Chosen First Name or Nickname (what do you like to be called)	Paul
Chosen Middle Name (optional)	(No response)
Chosen Last Name	Nieuwerburgh

7. Gender

This information is used for statistics, historical trends, and drafting top 40 finalist bios and is not an eligibility, evaluation or selection criteria.

Responses Selected:

Man

8. Pronouns

This information is used for statistics, historical trends and drafting top 40 finalist bios and is not an eligibility, evaluation or selection criteria.

He/Him/His

9. How do you identify yourself? Select all that apply.

This information is used for statistics and historical trends and is not an eligibility, evaluation or selection criteria.

Responses Selected:

White

10a. Parent/Guardian #1

First Name	Laura
Last Name	Veldkamp
Relationship to You	Mother
Occupation	Professor
Place of Employment	Columbia University
Department Name	Finance

10b. Parent/Guardian #2

Include your other primary parent/guardian here (if you have one), even if you do not share a household. Complete all fields below, unless you do not have a second parent/guardian.

First Name	Stijn
Last Name	Van Nieuwerburgh
Relationship to You	Father
Occupation	Professor
Place of Employment	Columbia University
Department Name	Finance

11. Are any of your parents, stepparents, guardians, grandparents, aunts, uncles, siblings and/or any other relatives or members of your household professional scientists or engineers, or do they work in a field related to STEM (medicine, dentistry, education, etc.), or work in academia?

If so, list their names, occupations and places of employment below, even if already stated above, even if you do not live in the same household, and even if you did not do research with them. **These connections are permitted but must be disclosed; failure to disclose could result in a failure to qualify.** (150 word limit)

Both parents are professors (academics) at Columbia University studying economics and finance. My belated grandfather was a scientist who worked at MIT

SCHOOL INFORMATION

12a. Current High School Name and College Board Code

Search for and select your high school below. If your school is not listed, type in "Add My School," and select "Add My School" from the list, then tell us more in the next question. Do not select a school that you do not attend.

333480,Bronx High School of Science,Bronx,NY,10468

12b. Add My School! (or confirm that your school is already listed)

- If your school is listed above, click "no" to this question.
- If your school is not listed above, click "yes" to this question and select "Add My School" as your school in the question above. After you mark this task as complete, our team will be notified to add your school. Check back in 2-3 business days, and you will find your school on the list. Please select your actual school and resubmit the task. We typically add 100+ schools to this list each year.

Would you like us to add your school?

No, my school is already on the list!

13. Where is your high school located?

I attend school in the United States or a U.S. territory.

14. Tell us more about your high school and your experience at your school.

Select all that apply.

Responses Selected:

My high school offers a research class.

15. Type of School:

Select all that apply.

Responses Selected:

Public

16. Current Coursework

In what classes are you currently enrolled? Enter up to eight (three are required). Grades are not needed.

1	AP Physics C
2	AP Literature
3	AP Microeconomics
4	Multivariable Calculus
5	Linear Algebra
6	App Development
7	Math Research
8	Orchestra

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 2a: Request Project Recommendation

Incomplete

Project Recommendation

View the [Project Recommendation Instructions](#) document to learn more about who should complete this form.

Click on the green "Request a Recommendation" button. Enter your recommender's name and email address. Your recommender will be able to see what you type into the message field. Click on the "mark as complete" green button to finish the task.

- Deadline: **Recommendations are due on November 6, 2025 at 8pm Eastern Time via this online system.** Give recommenders plenty of time.
- Each applicant must request one Project Recommendation, but may request up to two. *The option to request a second Project Recommendation will appear in this section after you have requested your first one.*
- Your recommender should receive an automated email. Recommendations are confidential. It is your responsibility to remind your recommenders to submit their recommendations. You will receive an automated email when each recommender submits, and you will be able to view the status at the bottom of this section.

Tips

- Sometimes entrants decide to swap/replace/reorder their requested recommenders. We do not recommend this unless you know for sure that an adult has NOT completed a recommendation on your behalf. If by chance the adult has already completed the recommendation, you will delete the content and we will not be able to retrieve it.
- **We highly recommend sharing the [Recommender FAQ](#) and the recommendation-specific instructions above with your adult recommenders so they know what to expect. You can find many troubleshooting tips for recommendations there as well as in our Applicant FAQ.**
- Mark this task as complete as soon as you have requested your recommendations and submit your application as soon as you are ready, even if your recommenders have not yet completed your recommendations. Recommenders will still have access to the form after you have done so.

Recommenders

Task 2b: Request Educator Recommendation

Incomplete

Educator Recommendation

View the [Educator Recommendation Instructions](#) to learn more about who should complete this form. The Educator Rec is NOT about the project--your recommender should be able to complete this well ahead of the deadline.

Click on the green "Request a Recommendation" button. Enter your recommender's name and email address. Your recommender will be able to see what you type into the message field. When you are done requesting recommendations, click on the "mark as complete" green button to finish the task.

- Deadline: **Recommendations are due on November 6, 2025 at 8pm Eastern Time via this online system.** Give recommenders plenty of time.
- Each applicant must request one Educator Recommendation but may request up to two. *The option to request a second Educator Recommendation will appear in this section after you have requested your first one.*
- Your recommender should receive an automated email. Recommendations are confidential. It is your responsibility to remind your recommenders to submit their recommendations. You will receive an automated

email when each recommender submits, and you will be able to view the status at the bottom of this section.

Tips

- Sometimes entrants decide to swap/replace/reorder their requested recommenders. Do not do this unless you know for sure that an adult has NOT completed a recommendation on your behalf. If by chance the adult has already completed the recommendation, you will delete the content and we will not be able to retrieve it.
- We highly recommend sharing the [Recommender FAQ](#) and the recommendation-specific instructions above with your adult recommenders so they know what to expect. You can find many troubleshooting tips for recommendations there as well as in our Applicant FAQ.
- Mark this task as complete as soon as you have requested your recommendations and submit your application as soon as you are ready, even if your recommenders have not yet completed your recommendations. You do not have to wait for your recommendations to mark this task complete OR to submit your application.

Recommenders

Task 2c: Request High School Report

Incomplete

Request your transcript on the day you open your application to allow your counselor plenty of time. Senior year grades are NOT needed – your counselor is able to complete this well ahead of the deadline. View the [High School Report Instructions](#) to learn more.

Click on the green "Request a Recommendation" button. Enter your recommender's name and email address. Your recommender will be able to see what you type into the message field. When you are done requesting recommendations, click on the "mark as complete" green button.

- **Deadline: Recommendations are due on November 6, 2025 at 8pm Eastern Time via this online system.** Give recommenders plenty of time.
- Each applicant must request one High School Report (transcript request). This report is NOT a traditional recommendation; counselors are only asked to upload your official high school transcript and answer a few questions about your school. If you have other transcripts from college courses, upload them yourself in the Test Score section, Task 9.
- Your recommender should receive an automated email. Recommendations are confidential. It is your responsibility to remind your recommenders to submit their recommendations. You will receive an automated email when each recommender submits, and you will be able to view the status at the bottom of this section.
- Homeschooled students may ask a parent or co-op leader to enter a transcript or grade summary.
- Transcripts from any college-level coursework should be uploaded by the student entrant in Task 9.

Tips

- Sometimes entrants decide to swap/replace/reorder their requested recommenders. We do not recommend this unless you know for sure that an adult has NOT completed a recommendation on your behalf. If by chance the adult has already completed the recommendation, you will delete the content and we will not be able to retrieve it.
- We highly recommend sharing the [Recommender FAQ](#) and the recommendation-specific instructions above with your adult recommenders so they know what to expect. You can find many troubleshooting tips for recommendations there as well as in our Applicant FAQ.
- Mark this task as complete as soon as you have requested your recommendations, and submit your application as soon as you are ready, even if your recommenders have not yet completed your recommendations – you do not have to wait for your recommendations to mark this task complete OR to submit your application.

Recommenders

Task 3: Rules Verification & Form Uploads

Completed - Jun 15 2025

Rules Wizard

Rules Wizard

This task will guide you to upload paperwork dependent on your research topic. Research that involves humans or human-related data, vertebrate animals, or human or vertebrate animal tissue and cell lines, PHBAs, prescription drugs, controlled substances, and/or hazardous materials must adhere to Regeneron STS rules. Researchers must obtain special permissions and approvals to work with these subjects.

- **Make sure any IRB and IACUC approval forms, risk assessments, wildlife permits, and any other paperwork is properly signed and completed, including any checkboxes that must be marked, and that any surveys and informed consent documents uploaded are blank.**
- Exempt studies will still need to upload documentation about the source of the tissue or data.
- *WARNING: This section contains many branched questions. Your response to one question will often cause a new, specific question to appear. If you change your answer within a series of questions, please take care to clear all answers in that subsection; otherwise, you may experience errors while trying to save the page. Some multiple-choice questions in the form do not have this option to clear your responses and may cause a question to briefly re-appear, but in these cases, the question should not cause an error and will disappear as soon as you have saved the page.*

PART I: RULES OVERVIEW & TIMELINE

START HERE: Read the [Regeneron Science Talent Search Official Rules 2026](#).

This is a required step in order for you to understand the eligibility requirements, rules regarding human and animal research, and more. The rules document contains a helpful checklist for all applicants. We strongly recommend that you review it. Here is a [list of common reasons](#) projects fail to qualify for Regeneron STS. If you have questions regarding the eligibility of your research email sts@societyforscience.org.

Responses Selected:

I verify that I have read the Rules and Entry Instructions 2026 document.

Share important dates in your research project timeline (short answer).

Regulated Research Institution definition: a professional research institution that is regularly inspected by the USDA and is licensed to use animals covered by the Animal Welfare Act. University labs, NIH, VA Medical Centers, CDC, and often pharmaceutical and biotech companies are Registered Research Institutions.

A. When did you start brainstorming your project? (75 words max)	I got the idea for my project when I realized there was a fundamental issue with the current paradigm of DNA binding whilst working on a training project.
B. When did you request permissions or preapprovals (if needed) from an IRB/IACUC/high school, etc.? (100 words max)	No permissions were needed for the public datasets that I used.
C. When did you begin collecting your data? If you collected different types/sets of data, please describe and share the start date for each type. (100 words max)	I used publicly available datasets
D. When did you conclude data collection? If still ongoing, please note and share when concluded for the sake of the research report submitted to STS. (75 words max)	N/A.
E. How did you get the data used in your research report? Possibilities could include (but are not limited to): self-collected through experimentation, provided by a mentor, from a public database, etc.? Private? Online? Self-collected? Please describe. (125 words max)	Provided by Public Databases
F. What type of data was utilized in your project? (survey responses, public database, collected via experimentation, etc.) (100 words max)	Public databases with experiments involving DNA. No human data was used.
G. Was this research conducted (in full or in part) at or under the supervision of a Regulated Research Institution? See above for the definition. (50 words max)	Yes, my training project was facilitated by Professot Bussemaker at Columbia University and while conducting my individual research I presented occasional progress updates for scientific rigor to Prof. Bussemaker.

PART II: HUMAN RESEARCH

A. There are many types of human studies; *please CHECK ALL THAT APPLY to your project* (we will ask about human tissue and cell lines in a later question).

Studies involving humans often require approval from an Institutional Review Board (IRB) before the start of experimentation. The IRB will determine if a student researcher's project is safe, and whether human participants should sign Informed Consent forms. Most entrants who require IRB approval will undergo this review (before experimentation) through the IRB of the University if research is at an RRI and at the school/affiliated fair level if conducted independently. Research conducted on high school students can usually undergo the IRB approval process through a high school-level or district-level IRB; research conducted through a university, hospital, etc. should undergo the IRB approval process at the institution with the greatest connection to their research subjects. If needed, visit this [guide to how schools and districts can form IRBs](#).

Responses Selected:

9. I did NOT work with humans or human data (if you worked with human cell lines/tissues, tell us in Part 3).

PART III: HUMAN TISSUE AND/OR CELL LINES

A. Does your project involve the use of human tissue or cell lines? Or images or data related to human tissue or cell lines?

Examples include epithelial cells, muscle cells, blood from blood banks, human breast milk, etc.

Responses Selected:

3. No

PART IV: FINAL VERIFICATION FOR HUMAN-RELATED PROJECTS

A. Verify your use of deidentified data:

1. I worked with de-identified data. (This means that there were no names with the data when it was given to you. If it did come with names, do not check this box.)

B. Inventions/Human Testing

Did you create anything that other humans helped test for you (software, video game, exercise routine, invention, etc.)? AND/OR

Did you ask humans to take ANY action or follow any instructions (complete a survey, provide any type of feedback, perform a dance, measure health data, etc.) for your project? AND/OR

Did you take original photos or videos of humans reacting to stimuli, performing certain tasks, etc. for the purposes of training an algorithm or any other use?

3. No

C. Working with Minors

Did you work with live humans under the age of 18 (or with tissues from humans under age 18)?

Not Applicable

D. Permission to Work with Human Audience

If you worked with live human populations through an institution (day care, memory care, hospital patients, etc.) please upload proof of permission to work with this audience. A letter from the institution with oversight over the population studied will suffice.

Note that this section is 2 pages. Click "Next" to proceed to the next page.

PART V: VERTEBRATE ANIMAL RESEARCH

A. Does your project involve live, non-human vertebrate animals or non-human vertebrate animal tissue or cell lines?

Many projects involving vertebrate animals, cephalopods (squid, octopi, etc) and/or cell lines and/or related data will require pre-approval by an IACUC. This approval process must take place through a Regulated Research institute (university, hospital, etc.) with the greatest connection to their research subjects.

For the purposes of this survey, the phrase "vertebrate animals" also includes cephalopods.

Select all that apply:

Responses Selected:

5. No, I did not work with non-human vertebrate animals

PART VI: RESEARCH INVOLVING POTENTIALLY HAZARDOUS BIOLOGICAL AGENTS (PHBAs)

A. Did you work with PHBAs, including microorganisms (including bacteria, viruses, viroids, prions, rickettsia, fungi and parasites) or recombinant DNA technologies, even as a small part of your study?

- This applies to projects that tangentially used PHBAs. For example, if you fed E.coli to C.elegans, this counts as a PHBA project and you should click yes.
- Projects involving blood, breast milk or other tissue of unknown origin or that did not come from a certified bank or RRI must select yes. This type of work must be conducted in a BSL-2 setting.
- All projects involving human or animal waste should select yes.
- All projects involving human or vertebrate animal tissues should select yes.

No

PART VII: HAZARDOUS MATERIALS & ACTIVITIES

A. Did you work with hazardous materials and/or activities such as chemicals, hazardous devices or inventions, weapons, drones, controlled fires and/or radiation?

No

PART VIII: PRESCRIPTION DRUGS AND CONTROLLED SUBSTANCES

A. Did you work with prescription drugs or controlled substances?

No

PART IX: ADDITIONAL PAPERWORK & INFORMATION

A. If you have not already been required to do so, please share what safety precautions, if any, were taken to conduct your research (100 words).

All publically available data; none collected myself

B. Environmental Impact (maximum 100 words)

Consider the impact of your research study on your surroundings. Describe what happened to your research subject post-experimentation. Describe (as applicable):

- What happened to the vertebrate/invertebrate animals after experimentation?
- The disposal of hazardous chemicals and substances or by-products of your research?

N/A

C. Optional Paperwork Description

If you wish to provide additional paperwork to show approval processes or continuation paperwork from your local science fair or Regeneron ISEF, etc. you may do so here. You may also upload any additional IACUC or IRB approvals, receipts or letters that document the origins of human or non-human vertebrate animal projects that you have and would like to share, especially if you were not specifically asked for this documentation in the form. Please describe what you are uploading and why in the text box and upload your documentation in the next upload section.

(No response)

D. Optional Paperwork File Upload

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 4: Science Research Description

In Progress - Last edited: Jun 16 2025

Science Research Description

Tell us about your research project! A project must have data and results to draft a research paper. You are asked to discuss your results both in the paper and in the essays within the application. Brief responses are acceptable; it is not always necessary to use the full word limit for each question. Clarify your contribution to the research report you are submitting and describe your research process. Be honest about all outside support.

NOTE: Use of special characters (specifically, emojis) may cause errors when attempting to save this page.

TITLE & CATEGORY

1. Project Category

Select the category that best fits your project. Review [this list of category descriptions](#). Unlike other competitions, prizes are not awarded by category; your category choice is only relevant for selecting the expertise necessary to review your project. If judges or evaluators feel additional expertise from another category is necessary to review your project, additional reviews will be assigned.

Cellular and Molecular Biology

2. Project Theme (optional)

Research submitted to Regeneron STS is often interdisciplinary. This information is primarily used for media purposes and identifying future trends in categories. *Please select all that apply.*

Responses Selected:

Computational Sciences

Genetics

Machine Learning

Microbiology

3. Project Title

Enter your project title. For any symbols, please write the name of the symbol in all capital letters (ALPHA, GAMMA, etc.) Otherwise, please use the AP-style Title Case. If you need help determining the Title Case for your project title, [use this website!](#)

Full Project Title:	Revolutionary Framework for Predicting In Vivo Transcriptional Regulation from In Vitro Protein-DNA Binding Models: A Breakthrough in Computational Gene Regulation
If the title of your project requires any special symbols or formatting (such as italics) please explain here:	(No response)
Short Project Title (100 characters or less, to be used in some publications):	Predicting In Vivo Transcription from In Vitro Binding: A Breakthrough in Gene Regulation Modeling

RESEARCH ENVIRONMENT

4. Research Environment

Where was the experimentation/research conducted?

Select all that apply.

Responses Selected:

College or University

5. Research Experience

Please check all that apply to this research experience:

"Mentor" below can mean a scientist, engineer, mathematician, etc. Mentorship can be formal or informal. "Research" can apply to science, mathematics, engineering, and invention.

Responses Selected:

Identified and contacted a mentor independent of any support

Worked in a lab with a research mentor

6. Research Payment Disclosures

Select all options that apply to this research experience in terms of payments. Per the Regeneron STS Official Rules, you are required to disclose any payments made by you or on your behalf for your research experience or training.

- Entrants who do not disclose involvement in a paid program will be disqualified.
- All options below are permitted and very common.
- It is important to credit any level of coaching received, including application, essay or competition coaching related to this application.
- If you have paid your mentor/coach at any point in the past, even if for a different project or circumstance, you should disclose this information.
- "I" in all cases below references you and/or anyone who made payments on your behalf.

Select all that apply:

Responses Selected:

None of the above, my research experience did not involve fees as described above

6a. Fees Paid

Program/Mentor/Coach Name (to whom were fees paid)	N/A
Tuition/Total Fees (not hourly fee or only initial registration fee; provide original price for the program before scholarships applied)	N/A
Explanation of Fees (residential program, meals, tuition, any scholarships received, etc.)	N/A
Notes	N/A
Website	N/A

7. Primary Research Location

Name of Research Location	Columbia University
Describe the Research Location (institution, university, park, school, etc.; please include name of lab)	University
Was the primary research location in the United States (or a U.S. territory)?	Yes
Research Location City	New York City
Research Location State	NY
Website	https://bussemakerlab.org/
General Contact Email	(No response)

8. Mentors

Please list your mentors (adults, undergraduate students or higher, with whom you met and worked in any way related to your research). You will be asked to provide their contact information in Task 10. These fields are optional; if you did not work with a mentor, you may skip them, but if you did, please be sure to complete.

	Prefix	First Name	Last Name	Institution (if different from above)	Job Title	Time spent with this mentor
Mentor #1	Prof.	Harmen	Bussemaker		Professor	6 months
Mentor #2						
Mentor #3						

9. How did you meet your mentor(s) and how involved were they in your research process? If you did not work with any mentors, you may skip or discuss your level of independence. (maximum 100 words)

I cold emailed him, he interviewed me, and accepted me into his lab for the purposes of a training project and other lab work unrelated to my Regeneron project

STUDENT CONTRIBUTION

10. How did you get the idea for your research? (maximum 250 words)

- Explain the development of your research question and/or engineering goals.
- What inspired you to conduct this research project? What is the story behind your topic? Was the project assigned to you?

The initial concept emerged during my training project when I recognized the fundamental disconnect between controlled in vitro binding measurements and complex in vivo gene regulation. Following extensive training in computational biology methods, I independently identified that despite decades of research, past researchers lack reliable methods to predict transcriptional regulation from protein-DNA binding data, with conventional models achieving only 15-20% accuracy.

11. What was the duration of the research? (maximum 75 words)

Explain the amount of time you spent on the research project that you have submitted, including start and end dates.

4 months: 7/2/2025 to 11/5/2025

12. If your research was conducted as part of a larger research project or group, explain how your work is independent of this larger project and/or where it fits into the project. (maximum 150 words)

- If there were other high school students in the group (currently, or any work completed on this project in past years), be specific about how your work was similar to and different from other students.
- It is important to be honest about your contribution. Context helps us understand and score your entry. We understand that students often face limitations when it comes to the freedom of selecting a research question, operating certain types of equipment, etc.

My work is independent of a larger research project and is not a continuation of the research from the lab where I conducted my training project.

13. What Did You Do?

Describe your contributions to each part of the research process in detail through questions 13a-f below. Attribute the support you received in each area of the research process and highlight what you claim as your own, original, unique contribution.

NOTE: This is the bulk of the application where you can tell us what you did, explain your thought process, and more! Elaborate beyond your research paper. You may click and drag at the corner of each box to expand as needed; please note allowable word counts to the side of each box below.

a. Developing / Initiating the purpose of the research
(200 word max)

After completing my training project in Professor Bussemaker's laboratory, I independently identified a fundamental bottleneck in computational biology: the inability to accurately predict in vivo transcriptional regulation from in vitro protein-DNA binding measurements. Despite decades of research and billions in funding, conventional models achieved only 15-20% accuracy in predicting transcription factor binding patterns in living cells.

My original contribution was recognizing that this limitation stemmed from flawed conceptual assumptions rather than technical inadequacies. The entire field treated transcription factors as independent competitors for binding sites, following reductionist paradigms that ignore cooperative interactions and emergent network properties.

I independently proposed developing a revolutionary framework that would model protein-DNA interactions as components of a quantum-like field where cooperative binding creates emergent behaviors. This paradigm shift required abandoning conventional computational biology approaches and developing entirely new mathematical frameworks inspired by statistical physics and tensor mathematics.

My unique insight was that biological networks follow thermodynamic principles invisible to traditional approaches, requiring methods that capture probabilistic superposition states rather than binary binding decisions. This represented a fundamental reconceptualization of transcriptional regulation, moving from describing individual protein behaviors to predicting

emergent network properties that govern gene expression patterns.

b. Designing the procedures (200 word max)

I independently designed a comprehensive computational pipeline integrating multiple genomic datasets through novel mathematical frameworks. My original methodological contributions included:

First, I developed tensor decomposition algorithms adapted from structural biology to model high-dimensional protein-DNA interaction spaces. This required creating new mathematical approaches for handling cooperative binding between multiple transcription factors simultaneously across millions of genomic sites.

Second, I designed Bayesian inference methods that capture uncertainty in binding predictions while incorporating thermodynamic constraints from protein structure data. My unique approach models proteins in probabilistic superposition states rather than binary binding configurations.

Third, I created integration protocols for combining SELEX-derived binding models with ChIP-seq occupancy data, ATAC-seq chromatin accessibility profiles, and RNA-seq expression measurements. My innovation was developing information-theoretic frameworks that preserve biological signal while filtering technical noise across diverse experimental platforms.

Fourth, I designed novel evaluation metrics based on thermodynamic principles rather than conventional binary accuracy measures. This required developing new statistical frameworks that assess prediction quality across probability distributions rather than point estimates.

My complete pipeline processes terabytes of genomic data through Columbia's high-performance computing cluster using optimized parallel algorithms I developed specifically for this application. All computational procedures were designed and implemented independently following my training project.

c. Implementing the procedure (including special techniques or the use of special equipment) (200 word max)	(No response)
d. Gathering / Recording data (200 word max)	(No response)
e. Analyzing data (200 word max)	(No response)
f. Formulating conclusions (200 word max)	(No response)

14. Describe any limitations of your research. (maximum 200 words)

Limitations are restrictions and constraints on the methodology of your study that can influence the findings of, and reliability of, your project. Examples include any financial barriers, access to equipment or materials, age restrictions, internet limitations and more.

(No response)

15. Did you use AI in this project or Research Report or application in any way? If so, describe how AI was utilized and what programs were used. (maximum 100 words)

I did not.

16. Are you submitting a published research paper as your Research Report (full or partial)?

Published research is permitted; it is not required. We only recommend submitting published research papers if the entrant is the first or sole author of the paper, and if a mentor will attest that the bulk of this work can be attributed to you.

Otherwise, submitting group work conducted with adults can confuse evaluators, and they will not be able to tell your contribution from the whole of the work. Please consider drafting a version of the paper that highlights your own contributions in these cases.

No, the work I am submitting is not published.

17. Additional Support Disclosure

Who else helped you? Indicate any other substantive guidance received, as well as any prior research involvement or training that helped you in conducting your work on this project.

This can include any meetings with or guidance from a scientist you may not consider a mentor, a parent or teacher suggestion that helped refine the project, a coaching service to learn research skills, application review or coaching, or anything outside of working with your mentor. (maximum 200 words)

During my training project, I also worked closely with a PhD student at the same lab as my mentor.

18. Required Conflict of Interest Disclosure

Do you need to report a real or perceived conflict of interest?

You are required to acknowledge any type of relationship between you, your family and your mentor(s) of all types.

Consider:

- Are you related to anyone who works in the lab, environment, department and/or institution in which you conducted your research?
- Are you related to anyone who knows your mentor (in any capacity)?
- Are you related to anyone who works at the same institution as your mentor?
- Do you have a familial, long-term or pre-existing relationship with anyone who mentored you or might be writing a recommendation for you?
- Do your parents know your mentors in any capacity--working or personal or casual relationship? Have they published research together? Do they work in the same department?
- Is your mentor a graduate student in one of your parental figures or another relative's lab or at their institution?
- Do you have a pre-existing relationship to Society for Science staff or Society for Science Trustees?
- Anything else of this nature?

This is permitted and students are not penalized if they are related to their mentors or have other long-standing connections. **However, it is imperative and required by the STS Official Rules to disclose this information.** Non-disclosure could lead to an entry's failure to qualify for competition.

Yes

18a. Describe the relationship(s), state the name(s) of individuals involved, and describe their involvement in the research project. (maximum 200 words)

Both of my parents work at Columbia University, but neither had any relationship with my mentor at any point nor connected me to my mentor. Neither of my parents helped me with my research project.

19. Research Report Similarities

Is there any reason your Research Report might get flagged in a plagiarism or AI detector? Matches can occur when methods are common protocols, or AI was used within the project. As long as the paper is cited properly and mentors are credited, this is acceptable. (maximum 150 words)

No.

20. Statement of Independence

Entrants frequently perform research that is similar to that of parents, mentors, relatives, friends and/or other high school students. This is expected, since science is a cumulative process, each finding builds on a previous one. The influence and assistance of those around you might be reflected in your responses, but in some cases might be difficult to articulate because their roles are a bit less defined. We would like to give you the opportunity to reassure the evaluation committee that while the above-mentioned influences may exist, the work you have submitted is your own and not that of a parent, mentor, relative, friend or any other person.

Failing to disclose similar or related research of which you are aware or failing to mention any person who has either closely or loosely guided you, and their relationship to you and your family, is a violation of our rules and the ethics statement.

Please explain any research or person that has influenced your work, has loosely advised you, has shared methodology with you, contributed to your research, etc. Provide as much supporting detail as possible to explain their influence or contribution. (maximum 200 words)

Here is a non-exhaustive list of examples: my mother works in this field (even if she did not help with your project, perhaps she shared some articles or you were inspired by some of her prior work); my father did not mentor me but he does research on X and because we talk about it a lot at the dinner table I was also interested in this topic; my teacher asks all of the research students to work on the same organism for budget reasons so our work might look similar but is different; my mentor published a paper on a similar topic which I read and it helped me a lot; etc.

(No response)

FUTURE PLANS & PERMISSIONS

21. Do you have any patents related to your STS research?

Select all that apply.

Responses Selected:

I do not have, nor do I plan to apply for any patents related to my research.

22. Do you have any patents unrelated to your STS research?

Select all that apply.

Responses Selected:

I do not have, nor do I plan to apply for any patents unrelated to my research.

23. Intellectual Property and Viewing Your Application

You are required to discuss your Regeneron STS entry with any scientists with whom you worked to confirm that they do not have concerns regarding the sharing of this intellectual property for the purposes of the competition (see the [Regeneron STS Official Rules](#) for the Society's Statement on Intellectual Property). In addition to sharing your application with our judges, evaluators and public relations teams, your application may also be shared with your mentors and recommenders.

Responses Selected:

I have confirmed that I have permission to share this intellectual property.

24. Individual Research Projects

Combining individual research projects into team projects to submit to other competitions is against STS Rules. An individual project must remain so through competition season (through June 2026).

Please verify that you do not have plans to combine the research submitted to the Regeneron Science Talent Search with another high school student for a different competition.

I DO NOT have plans to combine this research with another high school student for competition.

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 5: Research Report Upload

Incomplete

Research Report Upload

Instructions:

1. Review the [Research Report Guidelines](#) document.
 - The Research Report should detail your original project. Papers that only outline a review of current literature or detail a plan for a forthcoming project (without results) are not eligible.
 - We know it can be difficult for certain types of projects to consolidate findings into the 20-page limit. Please be assured that since this is required for all computer science entries (and across categories), our judges understand that you sometimes have to compromise on the amount of code, data, etc. you can include.
 - Evaluators and Judges are not permitted to click on links! Do not link to external websites (unless part of your bibliography).
2. Make sure every single one of your images/charts/graphs are properly cited per the new [Citation Guide](#).
 - Images/charts/tables/graphs/etc. that you created should also be cited per the guide.
 - Lack of citation or an incorrect citation could result in disqualification.
3. Upload your Research Report below.
4. Be sure to hit the green "Mark as Complete" button.

Don't forget:

- 4MB max (system might allow up to 5MB, but we request you only upload 4MB for evaluation purposes)
- Name your file "LASTNAME.FIRSTNAME.ZIPCODE" using your home address zip code.
- Do not submit a PDF created from an image file (via scan or photograph)

Task 6: Previous Research

Completed - Jun 16 2025

Previous Research

Tell us about your previous science research projects. If you do not have many previous research projects to share, that is totally ok; for many successful entrants each year, the research they submit to Regeneron STS is their first project!

NOTE: *Most text boxes can be expanded by dragging down the striped triangle in the the lower right corner.*

1. Individual Science Projects (optional)

List any individual research projects to which you have contributed during high school (not class projects or projects conducted with fellow high school students; projects conducted with teams of adults or college students only can be mentioned here). A project does not need to have been submitted to a competition to be listed. List projects in chronological order, starting with the most recent project on top. Check the box in the far-right column if this is the research project you are submitting to Regeneron STS.

	Start Date	End Date	Supervising scientist/mentors	Project Title	Competition/Awards (if any)	STS Project?
1.	7/1/2023	8/2/2023	Prof. Lerrel Pinto	Bidirectional VAE: a new model of robot learning in ultra-low data environments	Submitted to Terra NYC	☒
2.	6/28/2024	8/17/2024	Prof. Sidi Chen			☒
3.	7/1/2025	10/6/2025	Prof. Harmen Bussemaker	Revolutionary Framework for Predicting In Vivo Transcriptional Regulation from In Vitro Protein-DNA Binding Models: A Breakthrough in Computational Gene Regulation		☒
4.						☒
5.						☒

2. Team Science Projects

Have you ever completed research as a team with another high school student?

No

You are submitting that you have never conducted research as part of a student team project. Thanks for letting us know! Just to clarify, if you HAVE ever conducted research as part of a student team, you need to check YES and answer questions 2a-d. Even if you were a primary member of a team or conducted one portion of the research, it is still considered a part of that team project. Team research is not eligible for the Regeneron Science Talent Search, but we are curious about your past projects. If you have not conducted any team research, continue on to question 3.

3. Conference Presentations, Abstract Publications, Open-Source Pre-Prints (arXiv), Student-Level Publications, Paid Publications, or Future Submission Plans (optional)

List any conferences in which you have presented your work or plan to present your work in the next 3 months; instances where the abstract of your research has been published; publications in student-level, non-peer reviewed, or pay-for publication journals; and future publication plans (including articles submitted to journals, but not yet published, if any). (maximum 250 words)

N/A

4. Ph.D.-level Peer-reviewed Journal Publications

Are you listed as an author or coauthor on any scientific publications? (you may select more than one option)

Responses Selected:

No

5. Research Programs, Science Training Institutes, Summer Programs (optional)

- Select the number of research programs or science coaching programs that you would like to mention, and then forms will appear. You are required to submit any programs that even tangentially supported your STS experience.
- Search for your program of choice and add dates of attendance. Check the box to indicate if your STS project was conducted at any of these institutions.
- If a program you would like to list does **not** appear in the feed, we are happy to add it (see next question).
- Entrants are not required to participate in official research programs.

(No response)

6. Do you need to add a NEW research program to the drop-down list?

1. **FIRST, check the drop-down list of available programs in the search bar below under Program 1.** We often receive requests to add programs that are already on the list, because the student is looking for a slightly different version of the name (i.e. TS Scholars vs. Technology Science Scholars). Please double-check before submitting your request.
2. **Then, if the program you want to list is NOT an option, check the box next to "Please Add My Research Program" below, and complete the form that will appear.** After you Save & Close this Task, our team will be notified to add your program.
3. Check back in **2-3 business days**, and you will find your program in the feed. Please select your program, change this answer to "NO," and resubmit the Task. We typically add 100+ programs to this list each year.

NO, my research programs all appear on the list in the Programs section, or I do not wish to list any programs.

RESEARCH PROGRAM PARTICIPATION

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 7: Essay Questions

Essay Questions

The word limit is the maximum allowed; concise answers below this limit are encouraged. First or third-person responses are acceptable. Please be aware that special characters (especially emojis) can sometimes cause errors when saving the page. We always recommend drafting essays in a Word document as a backup!

1. Research Project “Layperson’s Summary” (maximum 200 words)

Summarize your project in layperson’s terms, while maintaining scientific accuracy. Your explanation should be easily understandable and include background, procedures, conclusions and relevance. This summary will aid readers, including evaluators, journalists, and the public.

(No response)

2. Project Benefits and Impact (maximum 200 words)

What benefits do you think your research will bring to the world, and/or to your field? What additional steps, and by whom, might be needed for this benefit to be realized?

(No response)

3. Your Potential as a Scientist, Mathematician or Engineer (maximum 200 words)

Address through specific and concrete examples what characteristics you have that best demonstrate your affinity and aptitude for becoming a future leader in STEM. Consider:

- What have you done that illustrates scientific aptitude, leadership, curiosity, inventiveness, and/or initiative?
- What are your other STEM-related interests besides your project?
- What do you plan to study in post-secondary education and what occupation do you plan to pursue? What do you hope to be doing 10 years from now?

(No response)

4. Greatest Accomplishment or Challenge (COMMON APP)

Please answer ONE of the prompts below from the Common App; we prefer that you think beyond your research project in this essay. (maximum 350 words)

A. Discuss an accomplishment, event, or realization that sparked a period of personal growth and a new understanding of yourself or others.

OR

B. The lessons we take from obstacles we encounter can be fundamental to later success. Recount a time when you faced a challenge, setback, or failure. How did it affect you, and what did you learn from the experience?

OR

C. Some students have a background, identity, interest, or talent that is so meaningful they believe their application would be incomplete without it. If this sounds like you, then please share your story.

(No response)

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 8: Activities, Interests and Awards

Activities, Interests and Awards

EXTRACURRICULAR ACTIVITIES, VOLUNTEERISM & EMPLOYMENT

We want to know more about you, your day-to-day life, how you spend your time, and why. Responsibilities inside and outside the home, like taking care of siblings and working part-time, are as important for us to learn about as your clubs and extracurricular activities. There are many ways for students to demonstrate leadership and commitment.

Please select an activity type and describe your involvement in up to seven activities (do not feel pressured to use all seven). If you do not see an exact match for your activity in the list, select "other." You should select the activities that take up most of your time and/or are most important to you.

1. *How many activities would you like to list?*

7

Activity #1

Activity Type:	STEM-related Club
Name of Activity (type of sport, name of club, etc):	Science Olympiad
Your Role/Leadership Roles:	Junior Year: Event Manager for Wind Power & Optics events Senior Year: Team Captain
How many years have you participated in this activity?	4
Explain the time commitment to this activity:	In previous years it was about 3 hrs/week from october through march. This year, as co-captain of the team, it's more like 6 hours per week.
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	(No response)

Activity #2

Activity Type:	STEM-related Club
Name of Activity (type of sport, name of club, etc):	Machine Learning Club
Your Role/Leadership Roles:	Member Sophomore year President Junior & Senior Year
How many years have you participated in this activity?	3
Explain the time commitment to this activity:	About 1 hr/week
Awards won/accomplishments related to this activity (100 words)	As president, I boosted the club's membership from about 7 people each week to over 25. I turned the club into a cultivator of innovation and creation of new tools and applications in Bronx Science.
Anything else you would like to share about this activity? (75 words maximum)	(No response)

Activity #3

Activity Type:	Volunteer Work
Name of Activity (type of sport, name of club, etc):	Street Squash Academic Volunteer
Your Role/Leadership Roles:	Community Tutor
How many years have you participated in this activity?	1
Explain the time commitment to this activity:	2 hours/week
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	I worked closely with 2 students who had recently immigrated from Cote d'Ivoire, using my French skills to make math and english language topics accessible to them where no one else could.

Activity #4

Activity Type:	(No response)
Name of Activity (type of sport, name of club, etc):	(No response)
Your Role/Leadership Roles:	(No response)
How many years have you participated in this activity?	(No response)
Explain the time commitment to this activity:	(No response)
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	(No response)

Activity #5

Activity Type:	(No response)
Name of Activity (type of sport, name of club, etc):	(No response)
Your Role/Leadership Roles:	(No response)
How many years have you participated in this activity?	(No response)
Explain the time commitment to this activity:	(No response)
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	(No response)

Activity #6

Activity Type:	(No response)
Name of Activity (type of sport, name of club, etc):	(No response)
Your Role/Leadership Roles:	(No response)
How many years have you participated in this activity?	(No response)
Explain the time commitment to this activity:	(No response)
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	(No response)

Activity #7

Activity Type:	(No response)
Name of Activity (type of sport, name of club, etc):	(No response)
Your Role/Leadership Roles:	(No response)
How many years have you participated in this activity?	(No response)
Explain the time commitment to this activity:	(No response)
Awards won/accomplishments related to this activity (100 words)	(No response)
Anything else you would like to share about this activity? (75 words maximum)	(No response)

2. Summer Activities

What did you do in the last 3 summers? Tell us about any jobs, activities, research, travel, family responsibilities, etc.
(maximum 200 words)

I did research internships at NYU, Yale, and Columbia respectively developing the projects outlined in Task 6.

3. Activity Checklist

Do you currently participate, or are you an alum, in any of the following organizations or activities? *Please check all that apply, even if already mentioned above.* This section is used to share statistics with partner organizations.

Responses Selected:

FIRST Robotics

Founded a non-profit organization

Model United Nations

National Honors Society

SCIENCE COMPETITIONS AND PROGRAMS

4. Have you participated in any of the following?

Select all that apply.

Responses Selected:

None of the above

AWARDS

5. List special recognitions, awards, honors, and scholarships from both school and community, if any.
(optional)

Include national or international honors received during your high school career not previously listed in the activities section above. (250 words maximum)

10 total Science Olympiad medals

INTERESTS

6. Share a fun or interesting fact that we might otherwise not know about you.

(75 words maximum)

(No response)

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 9: Test Scores (Optional)

Test Scores & Additional Transcripts (optional)

Sharing your standardized test scores is optional. If you choose not to share your test scores, this decision will not be held against you. If you would like to share your scores, enter them in this task, then upload evidence that supports your scores below (screenshots are fine). Superscoring is permitted. If you are superscoring a Composite test score, use the date of the most recent score in the "test date" column. Students who have transcripts from college courses may upload them at the end of this section. All high school transcripts must be submitted through the High School Report form by your counselor.

1. SAT Scores

Enter your total and section scores below. Your test, cross-test or subscores are not collected in the form, but will be visible (if you have them) in the score report you upload at the end of this section.

	Score	Test Date (MM/DD/YYYY)
Total Score (Composite)	1560	12/07/2024
Section Score: Reading and Writing	770	12/07/2024
Section Score: Math	790	12/07/2024

2. ACT Scores

List highest score in each category on any date.

	Score	Test Date (MM/DD/YYYY)
Math		
Science		
STEM		
English		
Reading		
Writing		
ELA		
TOTAL COMPOSITE SCORE		

3. Advanced Placement Course Test Scores

(Science, Math, Engineering, and Technology tests first) List top 8 only.

	Subject	Score	Test Date (MM/DD/YYYY)
AP Test #1	Chemistry	5	
AP Test #2	European History	5	
AP Test #3	Calculus	5	
AP Test #4	US History	5	
AP Test #5	Language & Composition	5	
AP Test #6	Computer Science	5	
AP Test #7	Physics 1	5	
AP Test #8	French		

4. International Baccalaureate Course Test Scores

(Science, Math, Engineering, and Technology tests first) List top 8 only.

	Subject	Score	Test Date (MM/DD/YYYY)
IB Test #1			
IB Test #2			
IB Test #3			
IB Test #4			
IB Test #5			
IB Test #6			
IB Test #7			
IB Test #8			

5. US Territories: Other Tests

For students attending accredited overseas secondary schools only. If you do not attend school in the US and take different tests than those listed above, please report the test names and your scores here.

Do not write in this box if you live in the US and/or took the tests listed above.

(No response)

6. Upload Score Reports

Please include all records for the scores you shared in this section. You are not required to upload official score reports. You may upload up to 4 files; if you have more than 4 files, please merge them. We ask that you remove extraneous pages about how to interpret scores. Should you need assistance, email sts@societyforscience.org.

Screenshots are acceptable!

7. Upload College Transcripts

While your high school counselor must upload your high school transcripts in the recommendation form in Task 2c, you may upload other unofficial transcripts from college coursework here.

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 10: Beyond the Project

In Progress - Last edited: Jun 16 2025

Beyond the Project

Information collected in this section is confidential and will not be shared with evaluators or judges. The Society will use this information for general record keeping, statistics and to contact your parents/guardians, teachers, principals, etc. in the event you are named a Scholar or Finalist.

STUDENT INFORMATION

1. Your Contact Information

Your Cell Phone or Primary Number	3322073566
Your Home or Family Phone Number (optional)	(No response)
Your preferred email address for long-term contact:	paul.nieuwerburgh@gmail.com

2. Your Mailing Address

We will send your entrant package and any scholar or finalist-related mailings to this address.

Do you live in the United States or in a US territory?	Yes
Street Address	410 Riverside Drive
Street Address 2 (optional)	(No response)
City	New York
State	NY
Zip Code	10025
Country	United States
Is this your permanent address? If you attend a residential school and provided the school address above, select "no" to this question.	Yes

3. Date of Birth (MM/DD/YYYY)

07/25/2008

4. T-shirt Size

(adult men's sizes)

L

5. Languages

Share your ability to speak, read, or write languages other than English.

	Language	Fluency
1	French	Advanced Speaker
2		
3		
4		
5		

5a. If you speak a language that is not listed in the drop-down menu above, please type it in here and share your level of fluency.

	Language	Fluency
Unlisted Language 1	Dutch	Fluent Speaker/Bilingual
Unlisted Language 2		

6. Public Name

Please tell us how you would like your name to appear in public materials relating to the Regeneron Science Talent Search (official bio if selected as a finalist, press releases, etc). We collected your legal and chosen names in Task 1 but understand that you might prefer a different variation of your name to be used for public materials (print program, media stories, etc). *Warning: This field often auto-populates from your web browser. Please make sure that your name appears here, not the name of another individual.*

Public First Name	Paul
Public Middle Name (optional)	(No response)
Public Last Name	Nieuwerburgh

FAMILY INFORMATION

7. Do you have any siblings? (optional)

Yes

7a. Please list the name(s) and age(s) of your sibling(s):

Henry, 15

8. Are any of your relatives former top 40 finalists or former top 300 semifinalists/scholars/honors group in the Science Talent Search (under Regeneron, Intel or Westinghouse sponsorship?)

No

9a. Tell us more about Parent/Guardian #1: **Laura Veldkamp**

Email Address	lv2405@columbia.edu
Do you share a primary mailing address with this Parent/Guardian?	Yes
Marital Status in Relation to Parent/Guardian #2 named below (optional)	Married

9b. Tell us more about Parent/Guardian #2 (optional): **Stijn Van Nieuwerburgh**

Email Address	sgv2110@columbia.edu
Do you share a primary mailing address with this Parent/Guardian?	Yes
Marital Status in Relation to Parent/Guardian #1 (name is above)	married

SCHOOL INFORMATION

10. Please provide additional information about your high school principal, and the teacher most closely associated with your application.

The teacher listed here will receive prizes if you are named a finalist. The Society for Science is only able to recognize one teacher per finalist or scholar.

Principal First Name	Rachel
Principal Last Name	Hoyle
Principal Email Address	mhoyle@schools.nyc.gov
Teacher (at your HS) First Name	Vladimir
Teacher (at your HS) Last Name	Shapovalov
Teacher (at your HS) Email Address	VShapovalov@schools.nyc.gov

11. High School Newspaper

Please complete this section if your school has a newspaper or newsletter. We will notify this source if you are named a scholar or finalist so that your school community can celebrate this achievement. (optional)

Name of High School Newspaper	The Science Survey
Email Contact at High School Newspaper	thesciencesurvey@gmail.com

PROJECT INFORMATION

12. Mentor Information

In a previous task, you shared the name(s) of your scientific mentors for your STS project. Please provide their contact information:

**If the names are not appearing correctly, please leave the mentor names blank, reference Task 4 and enter the contact information for the mentor(s) you listed in slots 1, 2, and/or 3.

Mentor/Supervisor Name 1	Harmen Bussemaker
Phone Number of Mentor/Supervising Adult 1	(No response)
Email Address of Mentor/Supervising Adult 1	(No response)
Mentor/Supervisor Name 2	
Phone Number of Mentor/Supervising Adult 2	(No response)
Email Address of Mentor/Supervising Adult 2	(No response)
Mentor/Supervisor Name 3	
Phone Number of Mentor/Supervising Adult 3	(No response)
Email Address of Mentor/Supervising Adult 3	(No response)

13. Who has been the most positive influence on your scientific endeavors?

If you are named a top 300 scholar, we will send a letter to this person and a copy of the scholar book, thanking them for encouraging and assisting you.

Most Influential Person Prefix (Dr., Ms., etc.)	(No response)
Most Influential Person First Name	(No response)
Most Influential Person Last Name	(No response)
Most Influential Person Address	(No response)
Most Influential Person Address 2	(No response)
Most Influential Person City	(No response)
Most Influential Person Country	(No response)
Most Influential Person E-mail Address	(No response)
How do you know this person?	(No response)

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, "edit."

Task 11: Optional Research Study

Completed - Jun 16 2025

Optional Research Study

Regeneron and the Society for Science are conducting some research on the types of students who participate in the Regeneron STS. Your participation in this research study is voluntary and will not affect how your application is assessed. You may choose not to participate. Your responses will be confidential and kept separate from the application material; they will not be shared with judges at any stage of competition or review. All data are stored in a password-protected electronic format. The questions in this part of the survey should take less than five minutes to complete. This survey is an optional step to submitting your application; however, if you choose to complete this task, you must answer all questions below.

1. How did you learn about the Regeneron Science Talent Search (STS) competition?

Responses Selected:

From a teacher or school administrator

From a summer program, research program or mentor

2. Other than STS, how many STEM competitions have you participated in over the past two years? Example competitions: ISEF, International BioGENEius Challenge, JSHS, A science fair within your school or region.

b. 1

3. Many careers are a combination of multiple subject matters. With that in mind, as of today, think of the career you would like to have in the future. Within which field would it typically be classified?

a. Science

4. How sure are you about the specific career you want in the future?

Scale of 1 to 10 (not at all sure to extremely sure)

Scale

6

5. The work you did on the project you used to enter STS most affected your future goals by

d. Helping you build skills that achieve your goals

6. As a result of working on the project you used to enter STS, how would you rate yourself as a skilled researcher?

Scale of 1 to 10 (not at all skilled to extremely skilled)

Scale

9

7. Did the chance to win prize money affect your decisions at all?

Responses Selected:

f. No, it didn't affect your decisions much at all

8. What are the top benefits related to research skills you hope to get from entering or competing in STS?

Responses Selected:

a. Analysis

c. Critical thinking

d. Problem definition

9. What potential college or career benefit of the options below inspired you most to enter STS?

a. Access to leaders in science, math, and engineering

10. What potential benefit do you most hope to get from STS regarding sharing your research with others?

a. Meeting peers with similar interests

11. Please share any additional comments you have about how your project and/or entering STS has influenced you:

(No response)

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task. You can edit this task by clicking on it, then on the three dots, then "edit."

Regeneron STS Ethics Statement (Required to Submit)

Completed - Jun 15 2025

Regeneron STS Ethics Statement

To sign and then submit your application, you must agree to all terms of the Regeneron STS Ethics Statement. Sign, then submit this task, then submit your application!

The Ethics Statement below speaks to your personal conduct as a young adult within your peer community as well as your conduct as a research scientist, mathematician, or engineer.

Scientific Research & Academic Integrity

Responses Selected:

I certify that all the information provided is correct and complete without omission to the best of my knowledge and I certify that the Research Report I am submitting is my own individual work, not that of a student team, nor does it represent the work of others.

I understand that I am responsible for all aspects of my submission's authenticity: the research, the Application, and all other documentation in the application process, as well as the display board and oral presentation if I am selected as a finalist. I attest that my Research Report and images within are properly credited and cited, and that I did not use AI tools to draft the paper or responses to application questions.

I attest all content in this submission is exclusively my work in substance and in presentation. I further understand that scientific fraud, misconduct, misrepresentation of work or attribution thereof, or violation of the rules and/or eligibility requirements may result in disqualification and forfeiture of any monetary awards and that the Society reserves the right in such cases to bar future participation in Society programs.

I have acknowledged all potential conflicts of interest, payment for research programs or classes related to my training, and parental/familial involvement and outside support related to my STS project.

Judging and Intellectual Property

Responses Selected:

I understand and accept that the judging and evaluation process used by Regeneron STS/ the Society is confidential and proprietary and that by submitting a Regeneron STS entry, I expressly agree to Regeneron STS/the Society's right to maintain the confidentiality of such process.

I agree to unconditionally accept the decision of the judges as final and binding in all matters related to the Regeneron STS program and understand that my application and Research Report will not be returned to me but shall become the sole property of Regeneron STS/the Society. I understand that Regeneron STS/the Society makes no warranties, representations or guarantees, express or implied, in fact or in law, with respect to the Regeneron STS program.

I expressly waive any rights to challenge, inspect, observe or otherwise obtain any information that constitutes the confidential and proprietary information of the Regeneron STS/the Society judging and evaluation process. I also agree to permit Regeneron STS/the Society to use all information contained in my application in any way it deems appropriate for publicity purposes.

I certify that I have not used AI tools, like ChatGPT, to construct the research report or responses to questions in the application.

Eligibility

Responses Selected:

I certify that I am at least 18 years of age, that I am an emancipated minor, or that I possess legal parental or guardian consent to submit the Application and participate in the Regeneron STS, and that I am fully able and competent to submit the Application and to abide by and comply with the Regeneron STS Rules and Entry Instructions. In any case, I certify that I am over the age of 13.

I certify that I have read and fully understood all rules and eligibility requirements found in the Regeneron STS Rules & Entry Instructions and that I have complied with all rules and meet the eligibility for submitting this Regeneron STS entry.

I understand that if selected as a top 40 finalist I am obligated to complete all necessary forms and megaform tasks as assigned by the Society for Science throughout the spring to prepare for my participation in Finals Week.

I understand that the Society reserves the right to remove me from the program and forfeit any awards and/or bar future participation in Society programs for scientific fraud; personal or scientific misconduct; misrepresentation of work or attribution thereof; violation of the Regeneron STS rules, eligibility requirements, or the Regeneron STS Ethics Statement; and/or conviction of a criminal offense.

Final Statement

I attest that all content in my Regeneron STS application adheres to the Ethics Statement above, and agree to the terms of the application.

Responses Selected:

Yes, I attest to the above statement.

Regeneron STS Applicant Signature Required

Use your mouse to sign the Ethics Statement.

A handwritten signature in black ink on a light gray background. The signature is cursive and reads "Paul Newburgh".

Once you finish a task, mark it "complete" by pressing the green button at the bottom of the screen. You will see a green check mark next to the task.

Once all boxes are checked and you've completed the signature field, it's time. You are now ready to submit your application! Be sure to hit the green "Submit button" that should appear to the left once you mark this task complete!